Class 1: Difference-in-Differences

In-class student worksheet

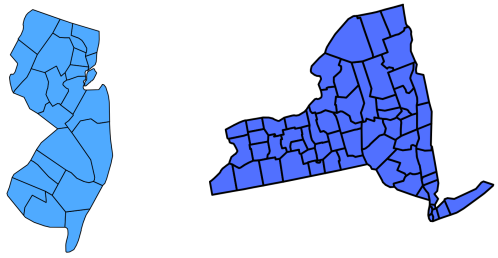
Name: _____ Date: _____

Learning Objectives:

By the end of today's class, we will be able to:

1. Quantify the effect of UI.
2. Compare correlation vs. causality.
3. Translate policy reforms into testable causal questions.
4. apply difference-in-differences methods to quantify the effect of a policy on the treated.
5. Group activity: DiD application with Stata or R

1. Why is a simple comparison insufficient?



UI =$628.29
19.3 weeks

UI =$553.97
17.9 weeks

Suppose State A has higher average UI benefits and longer average UI receipt than State B.

Quesiton: Can we conclude that higher benefits caused longer UI receipt? Give two other explanations for the difference between the states.

2. A policy change: New Jersey and Pennsylvania

In 1996, New Jersey temporarily made 13 additional weeks of UI benefits available. Pennsylvania did not introduce the same extension. We compare regular-benefit exhaustion rates before and after the extension.

A. Policy Question.

Question: Does extending UI benefits increase the share of claimants who exhaust their regular benefits?

B. Identify the design.

Treated state: _____ Comparison state: _____

What is the treatment? _____

What is the outcome? _____

C. Visualize with graphs.



D. Complete the table. All differences are in **percentage points**.

	Before	After	After − Before
New Jersey	49.9%	49.4%	
Pennsylvania	31.8%	29.0%	
NJ − PA			

E. Calculate the DiD estimate

$$\widehat{\tau} = (\text{NJ after} - \text{NJ before}) - (\text{PA after} - \text{PA before}) = _____$$

$$\widehat{\tau} = (\text{NJ} - \text{PA})_{after} - (\text{NJ} - \text{PA})_{before} = _____$$

3. Construct the missing counterfactual

A. Assume that, without the extension, NJ's exhaustion rate would have changed by the same amount as PA's. Complete:

$$\text{NJ after without extension} = \underbrace{_____}_{\text{NJ before}} + \underbrace{_____}_{\text{PA's change}} = _____\%.$$

B. On the axes below, plot both states' before/after outcomes. Add NJ's counterfactual after-period point. Label the estimated policy effect.



C. Interpret your estimate in one sentence. Explain why the estimated effect can be positive even though NJ's exhaustion rate fell.

4. When is the comparison credible?

5. Regression connection.

In the equation below, circle the coefficient that corresponds to your DiD estimate:

$$Y_{i,t} = \alpha + \beta NJ_i + \lambda T_t + \tau(NJ_i \times T_t) + \varepsilon_{i,t}.$$

$NJ_i = 1$ for NJ and 0 for PA; $T_t = 1$ after the extension and 0 before.



Table 1: Predicted Regular UI Exhaustion Rates with Regression

	Before	After	After − Before
NJ			
PA			
NJ − PA			